Ethan Schalnat

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OBJECTIVE

Mechanical engineering student seeking a full-time engineering position after graduation.

EDUCATION

Northern Arizona University — Flagstaff, AZ
Bachelor of Science, Mechanical Engineering — Jan 2023 – Dec 2026
Relevant Courses: Machine Design, Thermodynamics II, Solid Mechanics, Material Science, Statics, Dynamics, SolidWorks, Digital Circuit Design — GPA: 3.30/4.00

Red Mountain High School — Mesa, AZ
H.S. Diploma — May 2022
Relevant Studies: — GPA: 4.23/4.00
- PLTW accredited four-year engineering STEM pathway
- Led a team of 3 for stakeholder Rural Metro Fire, Inc. (ASU EPICS)
- Three years in robotics, designing and building robots suited to yearly VEX and FIRST robotics competitions

WORK EXPERIENCE

QuSpin, Inc.
Engineering Intern — May 2026 – Jul 2026
- Learned sensor manufacturing, calibration, RMAs, and customer support
- Designed and 3D-printed soldering, shield-alignment, and testing fixtures
- Performed QTFM (total field magnetometer) testing, including head loss, temperature and impact tests
- Redesigned the QTFM sections of the website, updating and merging sections for usability; Integrated sections into live site (WordPress)

224th Military Police Company — Glendale, AZ
Enlisted – Corporal (CPL) — Nov 2021 – Present
- Lead a team of 4 soldiers, managing accountability and readiness
- Conduct strict inventory management and inspections
- Deployed to Joint Readiness Training Center as a Vehicle Gunner
- Qualified as an expert marksman with the M4, M17, M249

PROJECT EXPERIENCE

Senior Capstone: Manufacturing and Test Engineer — Jan 2026 – Present
- Improving Ponderosa High School's off-grid renewable system, including PV production and storage and an interactive bicycle generator
- Develop the bike electrical diagram; designed for applicable codes
- Performed engineering analysis and calculations for bicycle generator

FLAMES Design Project: Team Leader of 5 — Aug 2025 – Dec 2025
- Worked with client Dr. Eilaghi
- Designed a prototype to automatically detect and suppress engine fires
- Performed extensive cost analysis including TVM and TBL
- Gained experience with CAD in Solidworks, XR modeling in eDrawings and Blender, and presenting to a board of engineers

SKILLS

Software:	CAD:	Autodesk Inventor, Fusion, SolidWorks, Blender
	Math and Coding:	MATLAB, C++, Python 3, Claude Code
	Electrical Systems:	DipTrace, LTSpice OrCAD, MSO28, Quartus II
Hardware:		Sensor calibration, soldering, 3D printing, jig and fixture design
Languages:		English (Fluent), Spanish (Basic)
Clubs:		ASME NAU Chapter, Renewable Energy Club

LEADERSHIP

Team Lead, FLAMES, Northern Arizona University — Aug 2025 – Dec 2025
Squad Leader, 2nd PLT 224th MP Co. — Nov 2023 – Present
Competition Lead, VEX Robotics — Aug 2020 – May 2022